ARISE

All-in-one middleware for industrial human-robot-interaction

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D2.1 Proof of Concept proposal

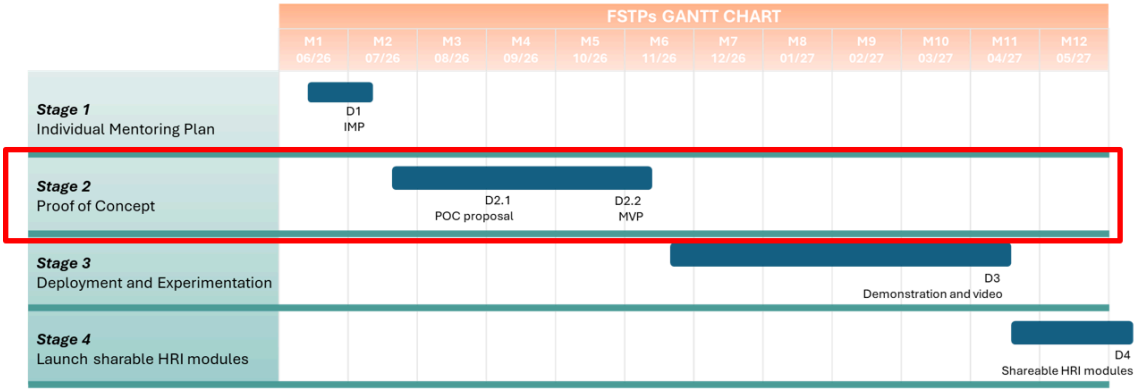


Figure 1: GANTT of the ARISE 12-month Support Programme.

Before starting

In each section, the corresponding instructions in italics might help to complete them. In some cases, open text is required to describe the activities and expected outcomes, in other cases tables are requested to be filled.

Please, use Exo 2 font with a size dimension of 10.

To describe a code, please follow the example in the following:

```
$ ls -l
```

PROJECT IDENTIFICATION	
Project title	**Posture-Aware Cobot Assistance in Packaging**
Project Acronym	**POSTURE**
Contact person	Beatriz Zambrano Serrano, beatriz@metamedicsvr.com, MetaMedics VR SL (MMVR)
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0. Recommendations

Stage	Recommendations	Response and actions taken	Mitigated
0-4	Specify the recommendation category (Technical, Ethical, Business, etc.) and the priority level (Critical, Preferred, Optional).	Reply outlining the actions taken and indicating the relevant sections in the IMP.	Yes/No
1	Document the data models to be implemented, including structure, semantics, and alignment with NGSI-LD.	Added the logic in page 11	
1	Provide a drafted software architecture, describing modules, interrelations, and data flows.	Added a logic flow and chart pages 13-15, and the annexes.	
1	Add details on the release strategy of the reusable module (e.g., repository location, licensing, documentation, tutorials, long-term maintenance)	Added throughout, we already have tutorials for subcomponents and these are WIP, so we will update them. Repositories will start getting compiled in D3 so we have them early for others to reuse. MMVR and UA will devote energies to maintenance and combined commercialization with UA providing robotics-specific expertise.	
1	Expand the KPI set, especially business KPIs, and refine them with greater granularity to better capture adoption, value creation, and exploitation potential.	Added KPIs in pages 20-22	

1 PoC Overview

1) Detail the objective of the PoC and the use-case considered.

- The problem at Pepri is that operators on the rework and packaging line manually hold, rotate and position blow-moulded parts during high-frequency cycles.
- This drives sustained wrist, neck and shoulder strain, and has generated operator complaints. Actually the workers believe the robot is holding them down, making the whole process more inefficient than if they were working without the robots.
- The PoC targets a specific station, reducing musculoskeletal risk on the strain-heavy holding and rotation tasks while keeping the operator fully in control of the pace and the process. We want to measuring data, and improve workers conditions while improving the factory throughput.
- Additionally, our focus is to be able to publish "ErgoAssist skill" so it can be reused, following the ARISE philosophy.
- Our focus is **accessibility**, so that *all* the workers are comfortable and safe when working with the robots, enhancing their conditions (bringing betterness and improved conditions) while **reskilling** the operators through the tutorials and onboarding XR training, so that more operators can comfortably work with the robot.

2) Broadly describe the PoC, indicating the technological approach, input used and tangible outputs.

- Instead of the operator adapting to the workstation, the workstation will adapt to them. A Universal robot collaborative robot presents and holds each part at the operator's neutral-posture position, and a height/tilt-adjustable fixture (the post-processing working table) can be changed per-operator profile. Given that we have seen a diversity in operators (men/women, left handed/right handed, weight, heights etc) so of course, one particular angle may not be comfortable for someone else.
- The operator will retain control through a touch-screen HMI and can pause, nudge or resume at any point without losing process context.
- The main changes and commands the operator will be able to control are: (present-at-angle, hold-and-rotate, yield-to-human on pause/nudge/resume rather than full E-stop and reteach). These are being validated with the advisory group.
- The hardware and interaction would be as follows: UR cobot plus adjustable fixture, a touch-screen panel at the cell, and a non-intrusive posture-sensing camera pipeline.
- It was intended that the operator would interact through the HMI and voice. However, the *voice* is currently an issue because with the noise from the factory saying numbers and having an AI understand these numbers has not currently been easy. Thereby, after speaking with our mentor and understanding that In a blow-moulding plant (>80 dBA), speech recognition without specialized directional hardware will be unreliable for operators, the touchscreen HMI with physical glove-friendly controls will be prioritized as the primary interface.

- **Inputs**: include operator profile, SKU parameters, live pose signals.
- **Outputs**: presented/held parts at ergonomic angles, plus a FIWARE dashboard logging ergonomics proxies, cycle time, safety events and first-pass yield, with automated alerts and reports.

3) Detail what is included/excluded from the trial. List assumptions, boundaries, and limitations.

- **In scope for the PoC:** one SKU, one shift, one Pepri station, validated with ≥8 operators. The flow includes an operator sits/stands at their profile-set station, the robot presents parts at their neutral posture, and they confirm quality or nudge the robot as needed, the same rework task with the strain removed.

- **Assumptions** are: Pepri confirms the SKU, operator pool and HMI target hardware by M2; the UR3 and station are available for integration.

- For this, multi-SKU and multi-shift scale-up are out of scope (deferred to the TRL-7 MVP, so the PoC first proves ergonomic and safety gains on one controlled case before adding variability).
- We are also keeping out of scope the digital-twin A/B testing and heterogeneous ERP/MES connectors.
- The PoC focuses on the different parts and its integration into a system: validated operator presets across body sizes and PPE, calibrated posture-proxy pipeline against RULA/REBA, and the interface-contract stability needed before scaling to three shifts and multiple SKUs.

At the current stage, the POSTURE PoC has demonstrated **functional communication and control** for the microcontroller-based ergonomic table and for the Universal Robot through ROS 2 Vulcanexus. The web-camera connection has also been tested.

The main unresolved technical issue, and the highest technical priority, is the stability and compatibility of the camera/ROS4HRI perception pipeline, particularly face and 6D-frame detection, which can crash the system. The table still requires the ROS 2 implementation that converts sensor voltage into distance and connects that value to the posture-related function. The HMI/DHMI–Vulcanexus contract is defined preliminarily: voice commands are interpreted by the HMI/DHMI and validated high-level commands are sent to ROS 2 to activate system modules.

FIWARE/NGSI-LD integration remains planned and has not yet been experimentally validated, which is then the second highest technical priority, for D2.2. Therefore, the present results confirm subsystem feasibility, while complete end-to-end ergonomic, HRI, safety and business validation remains pending, and will be solved in the next deliverable.

2. PoC Report

2.1 Performed activities

WP1 - Project Management, Safety and Quality	Duration: M1–M12
Objectives	Ensure coordinated execution across all partners; manage scope, schedule, budget, risks, ethics compliance, GDPR, safety standards, and quality gates throughout the full project lifecycle.
Summary of what has been done	Governance cadence established (T1.1): bi-weekly sprints and steering meetings running. Data protection framework drafted, DPIA, operator consent procedures and pseudonymisation approach defined (T1.3). Safety-standards mapping against ISO 12100, ISO 10218-1/-2 and ISO/TS 15066 underway (T1.2, WIP). Reporting gates (T1.4) tracked toward the PoC review.
Related to specific tasks	**T1.1 – Governance and Cadence (M1–M12)** Ongoing with meetings for technical sprints between tech partners, and monthly meetings with the end user. **T2 –Safety and Standards (M1–M12):** We are working with the end user to ensure everything developed is according to standards, health and safety standards. **T1.3 – Data Protection and Ethics (M1–M12)**: First initial ethics assessment analyzed and assessed. Adapted the project with recommendations around, operator consent procedures, pseudonymization, and more. Next to do will be the Ethics workshop. **T1.4 – Reporting (M1–M12):** Ongoing. IMP sent and now this D2.1 deliverable + video. https://www.youtube.com/watch?v=nhCK4brlUWo

WP2 -**Ethics, Human-Centricity and Inclusivity**	Duration: M1–M12
Objectives	Embed ethics-by-design and human-centred design principles across all project phases; ensure GDPR compliance, informed consent, and operator wellbeing; establish confirmed baselines and site readiness; guarantee

	inclusive access for diverse worker profiles; and monitor social and environmental impact
Summary of what has been done	Site visit completed at Pepri: on-floor observations and operator interviews carried out, and baseline ergonomic data gathered on the target rework/packaging tasks (T2.2). Cross-shift Operator Advisory Group formed, 8 operators recruited. This group is our human-in-the-loop channel: they are actively reviewing the HMI now and their feedback feeds directly into design (T2.3, ongoing). Formal Requirements & KPI specification is WIP, being finalised from the site-visit data.
Tasks	**T2.1 – Ethics and Data Protection Framework (M1–M2)** Developed the DPIA, operator consent forms, pseudonymisation procedures. Took in recommendations from the Initial Ethics Assessment. **T2.2 –Requirements, Baseline and Site Readiness (M1–M3)** Completed operator interviews and on-floor observations at the Pepri station, at the site visit. Also confirmed KPI and baseline specification for direct measurement, and hardware needs. **T2.3 – Participatory Design and Operator Co-Creation (M1–M4)** Established the cross-shift Operator Advisory Group, and ran meetings and advisory. Ongoing wireframes and HMI requirements validation with advisory group members, for continuous refinement and improvement, in alignment with the Initial Ethics Assessment.

WP3 -**Virtual Prototype and System Development**	Duration: M1–M8
Objectives	Develop and validate all technical components of the POSTURE solution in a controlled environment before live deployment, including the adaptive fixture, the ErgoAssist ROS 2 reusable skill, the HMI and XR training modules, and the FIWARE monitoring application.
Summary of what has been done	Framework technology research complete.

Tasks	**T3.1 –Interface Contract and System Architecture (M1–M2)** The ROS 2/Vulcanexus orchestration layer has been established and the principal data paths validated at component level: Arduino/sensors/relays → ROS 2 node → fixture functions; camera → camera node; UR driver → robot controller; HMI/DHMI → high-level ROS 2 commands. The architecture deliberately separates high-level operator intent from low-level hardware control, so the HMI never issues joint-level commands. The full formal interface contract — message definitions, actions, services, robot states (READY, MOVING, YIELDED, EXCEPTION, CYCLE_COMPLETE), timeouts and exception behaviour — is being consolidated and will be frozen before final multi-module integration. Status: architecture defined and component paths tested; formal contract WIP. **T3.2 –Adaptive Fixture Design and Prototype (M1–M4):** The height/tilt-adjustable fixture has been tested as a complete electromechanical subsystem. Verified functions: Arduino ↔ ROS 2/Vulcanexus communication, distance-sensor voltage acquisition, relay activation/deactivation, and motor movement in both directions. The remaining task is the ROS 2 function that converts the measured sensor voltage into a calibrated distance value and links it to the posture-aware logic; the calibration curve, units, valid operating range and error behaviour will be documented before the ergonomic KPI is evaluated. Status: functional subsystem test complete; voltage-to-distance conversion and posture linkage in progress. **T3.3 – ErgoAssist ROS 2 Reusable Skill (M2–M8):** The basic robot-control path is functional: the Universal Robot was driven through the UR driver connected to Vulcanexus, using both direct terminal commands and scripted execution, both operating correctly. This is a successful subsystem and integration test rather than final proof of the ergonomic use case. Remaining work: wrap the tested commands in formal ErgoAssist actions (present, hold, rotate, yield-to-human), validate state transitions and pause/yield/resume behaviour, define exception handling and complete collaborative-safety validation under the intended operating conditions. Status: robot communication validated; formal skill and safety/exception flows in progress. **T3.4 – HMI, Voice Interface and XR Training Modules (M1–M6):** The inclusive touch-screen HMI has been built and is in active feedback rounds with the Operator Advisory Group. The preliminary HMI/DHMI contract separates voice processing from robot control: voice commands are interpreted at the HMI/DHMI level, which then sends a validated high-level ROS 2 command to the relevant module, with a visible manual fallback if recognition fails. The five Unity XR micro-lessons are in progress: the base project is in place, but final content is held until PoC delivery so the training reflects the exact validated workflow. Status: HMI built and in feedback; ROS 2 command contract and XR lessons in

progress.

T3.5 – FIWARE Monitoring Application (M1–M6):
The NGSI-LD context layer (Orion-LD) has been designed but not yet tested experimentally; no Orion-LD component test has been performed, so no dashboard or ROS 2–Orion-LD bridge is claimed as validated at this stage.
The planned flow is: ROS 2 publishes robot, table, sensor and HMI events → an integration bridge maps selected events to NGSI-LD entities → Orion-LD stores and exposes them → a dashboard presents cycle, ergonomic, safety and exception information. Initial entities defined: Robot/RobotStation, ErgonomicTable, DistanceSensor, ProductionCycle, OperatorPosture/ErgonomicObservation, SafetyEvent and HMICommand, each with a stable identifier, timestamp, source, value/unit and relationships. Only derived, non-identifying data will reach this layer — raw camera frames stay at the edge.
Status: data models and architecture defined; first component test pending.

T3.6 – Posture Sensing and Calibration (M2–M6):
The RULA/REBA assessment method has been researched and the key questions digitised as the basis for the posture-proxy pipeline. The web camera has been connected to Vulcanexus and tested at connection level. Higher-level perception is not yet stable: face detection and 6D-frame detection can crash the system, and a library compatibility problem in the ROS4HRI environment remains unresolved. Next steps: isolate the failing dependency, pin compatible ROS 2/Vulcanexus and ROS4HRI versions, test the camera pipeline without face detection, test frame detection independently, then reintroduce components incrementally with process monitoring and controlled fallback.
Status: RULA/REBA digitised and camera connected; perception pipeline WIP, ROS4HRI compatibility unresolved.

2.2 Alignment with IMP

The activities remain aligned with the IMP and the original POSTURE objective: to develop a human-centred ergonomic workstation in which the robot and table adapt to the operator. The work has followed an incremental validation strategy, beginning with local component tests and then progressing to the robot and interaction subsystems.

The completed tests provide an initial technical foundation for the PoC. In particular, the Arduino/table subsystem and the UR/Vulcanexus subsystem have reached the functional test stage. The factory visit and operator surveys also maintain the human-centred approach required by the IMP. The advisory group has been created and asked, and baseline data has been recorded.. The main deviation from the planned sequence is that the camera/ROS4HRI pipeline has encountered compatibility and stability problems, while the FIWARE/NGSI-LD integration has not yet been tested. These issues do not change the use case, but they require additional integration and troubleshooting work, and will be the prioritized next step.

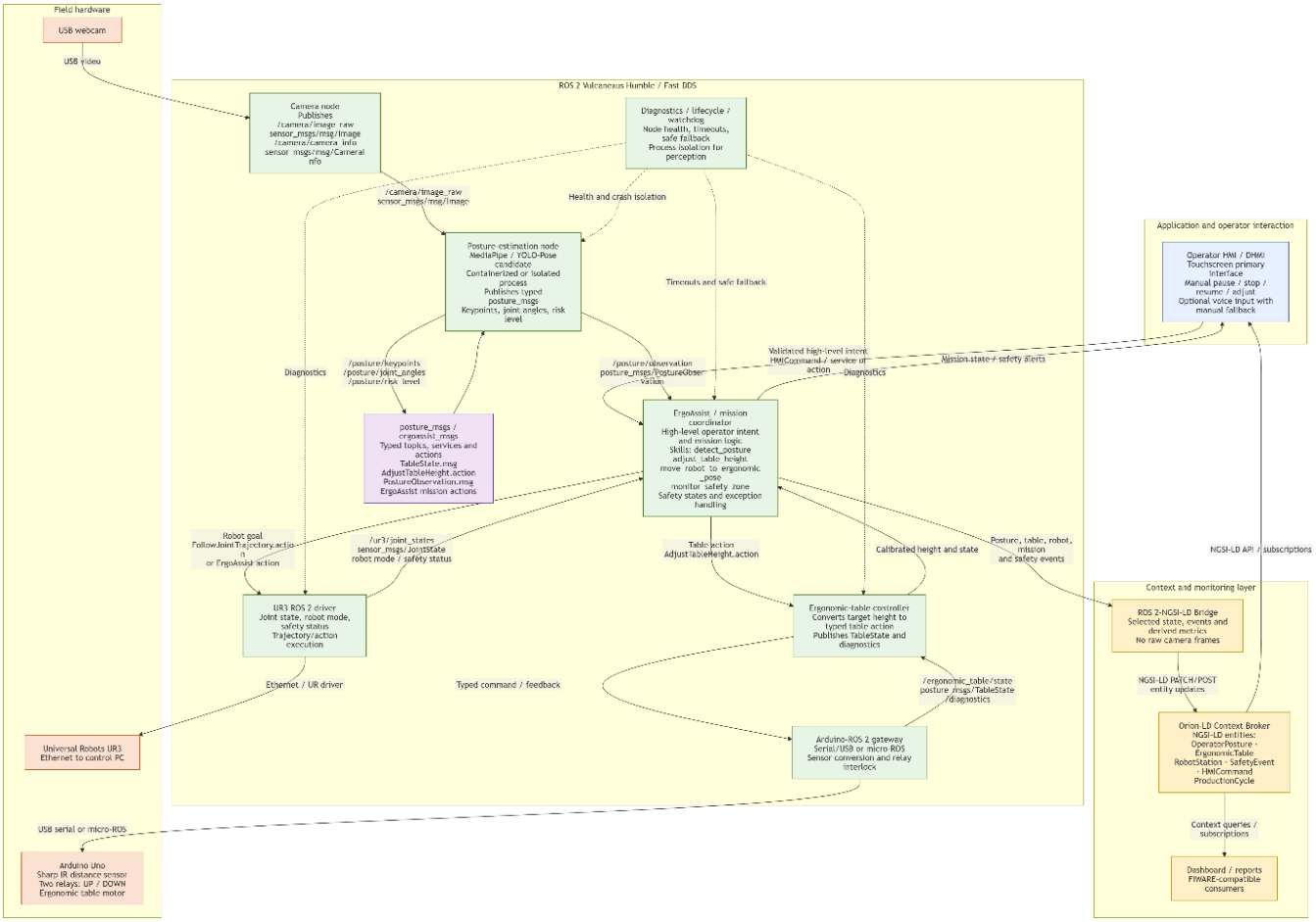
The idea is that already in the next deliverable, these are all mostly solved, so that we go from TRL 4 to TRL 5, and slowly move to TRL7, in m10.

We would leave the remaining 2 months of the project, m10-m12, to condense the impact, case study and other tools (such as tutorials) for the reusable models, and personalized commercialization actions.

2.3 Technology stack

The POSTURE PoC is implemented around ROS 2 Vulcanexus Humble (Fast DDS) as the middleware and orchestration layer, with FIWARE Orion-LD as the NGSI-LD context broker for semantic representation of the workstation state. Figure X presents the formal software architecture, showing all ROS 2 nodes, message types and interface boundaries between operator interaction, middleware, context management and field hardware.

The architecture is organised in four layers, as can be seen in this Figure.



The current technology stack is centred on ROS 2 Vulcanexus as the middleware and orchestration layer. The principal components are:

Layer	Component	Current status
Robot control	Universal Robot, UR driver, robot controller	Communication tested successfully through terminal commands and scripts
Middleware	ROS 2 Vulcanexus	Configured and used for Arduino, camera and UR tests
Table control	Arduino, distance sensor, relays and motorised ergonomic table	Component and functional tests completed
Perception	Web camera, face detection and 6D-frame detection	Camera connection tested; higher-level pipeline unstable
HRI	ROS4HRI libraries/conventions	Evaluated; compatibility problems remain unresolved
Operator interface	HMI/DHMI with preliminary voice-command contract	Operator feedback collected; final ROS 2 command contract pending
Context/data layer	FIWARE Orion-LD and NGSI-LD	Planned, not yet tested in the current PoC stage

The intended logical architecture is as follows:

- The Arduino node reads the distance sensor and controls relay outputs through ROS 2 interfaces.
- The table-control logic converts operator or posture-related requests into relay commands.
- The UR driver provides the ROS 2 connection to the Universal Robot.
- ErgoAssist will provide higher-level robot actions, including presentation, holding, rotation and yielding control to the operator.
- The camera node provides perception data. Face and 6D-frame processing remain isolated until their compatibility and crash behaviour are resolved.
- The HMI/DHMI receives operator input, including optional voice commands, and publishes high-level commands to ROS 2.
- The FIWARE bridge will later transform selected ROS 2 events into NGSI-LD entities for monitoring and analytics.

The architecture deliberately separates high-level operator intent from low-level hardware control. The HMI should not send joint-level commands. Similarly, speech recognition should be encapsulated in the HMI/DHMI and expose only validated high-level intents to ROS 2.

Figure 1 represents this four-layer model. Operator interaction occurs at the Application/HMI layer, which exchanges context with the FIWARE layer through the NGSI-LD API. The ROS 2 middleware layer (Vulcanexus/ROS4HRI) hosts the Mission Coordinator and the perception, actuation and visualization nodes, which interface with the physical devices in the field layer. The interface boundaries between layers are marked explicitly, and the ROS 2 nodes are annotated with the topics they publish and subscribe to.

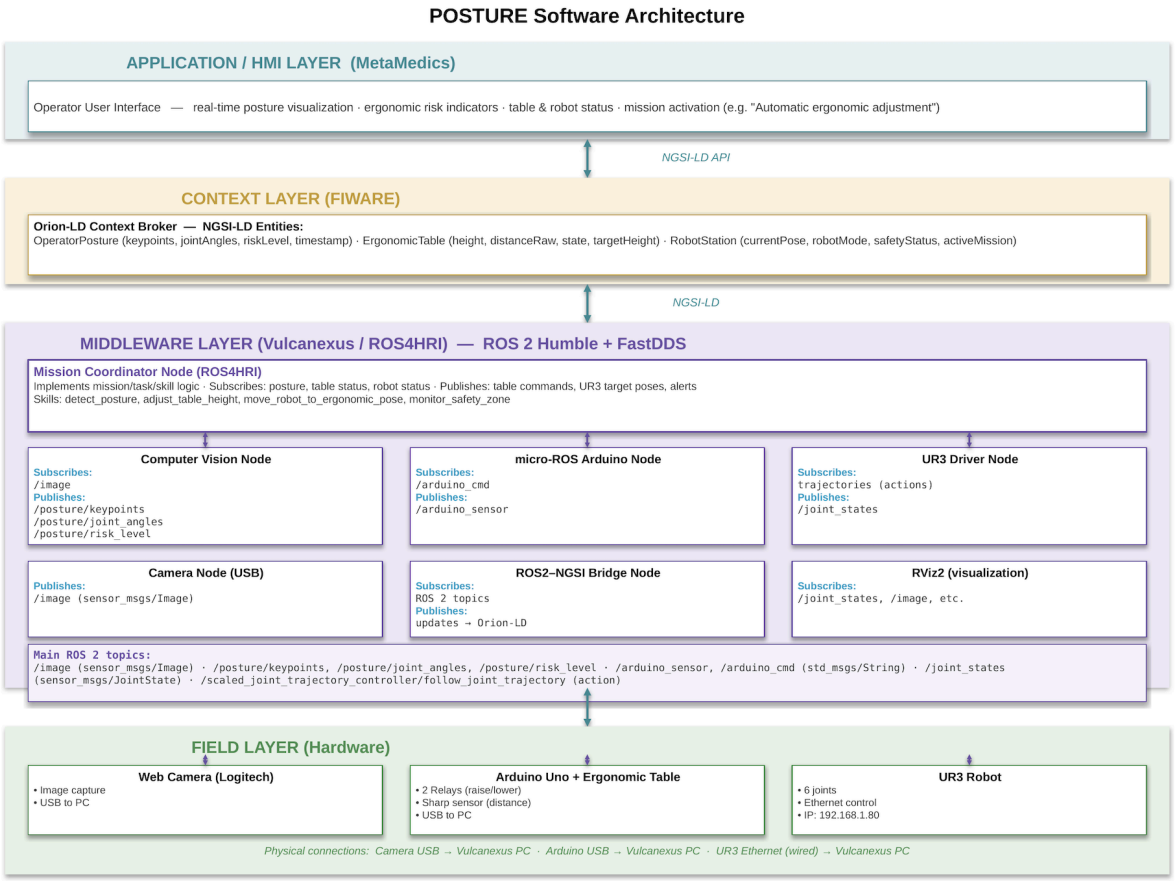


Figure 1: Software Architecture.

Middleware and ROS 2 layer

Vulcanexus Humble Desktop provides the ROS 2 Humble environment and Fast DDS-based middleware for the current implementation. Nodes may run on one or more processes on the main PC and communicate through topics, services and actions. The following interfaces define the intended architecture; exact package names and message definitions will be frozen in the interface contract.

Node/module	Main inputs	Main outputs	Status
UR3 driver node	Motion requests, trajectory goals, configuration commands	/ur3/joint_states, /ur3/robot_mode, /ur3/safety_status, action feedback/results	Communication and command execution tested; full high-level action integration pending

Arduino–ROS 2 gateway	Serial/USB data from Arduino; /ergonomic_table/command	/ergonomic_table/distance, table state and diagnostics	Validated for sensor voltage and relay/motor control; calibrated distance output pending
Camera node	USB camera stream	/camera/image_raw, optionally /camera/camera_info	Connection tested
Posture-estimation node	/camera/image_raw	/posture/keypoints, /posture/joint_angles, /posture/risk_level	In development; crashes observed in face/6D frame workflow
ROS4HRI mission coordinator	Posture, table and robot state	Table commands, UR3 goals, mission events, alerts	Architecture defined; implementation and compatibility validation pending
ROS 2–NGSI-LD bridge	ROS 2 posture, table, robot and mission data	NGSI-LD updates in Orion-LD	Planned; no Orion-LD component test completed yet

The intended table interfaces are /ergonomic_table/distance and /ergonomic_table/command. The distance topic will publish the calibrated table distance/height as std_msgs/Float32 or a dedicated posture_msgs/TableState message. The command topic will receive high-level values such as UP, DOWN and STOP, or an equivalent typed enumeration. The Arduino gateway translates these commands to relay outputs.

For robot control, the UR3 driver publishes joint states, robot mode and safety state, and offers services/actions for commands and trajectories. The exact implementation may use FollowJointTrajectory or a project-specific action for target poses, depending on the selected UR ROS 2 driver interface. The mission coordinator must use only validated high-level commands and must receive status/feedback before issuing subsequent commands.

FIWARE and NGSI-LD context layer

The planned context layer uses Orion-LD as an NGSI-LD-compatible Context Broker. It provides a shared semantic representation of the workstation for the HMI, analytics, logging and future integration with FIWARE-compatible systems. This layer has been designed but has **not yet been tested experimentally**. .

The initial NGSI-LD entities are as follows:

Entity	Example attributes	Purpose
OperatorPosture	keypoints, jointAngles, riskLevel, observedAt	Represent derived, non-identifying posture information

ErgonomicTable	height, distanceRaw, state, targetHeight, observedAt	Represent table measurement, motion and target configuration
RobotStation	currentPose, robotMode, safetyStatus, activeMission, observedAt	Represent robot and station operational context
ProductionCycle	cycleId, SKU, startTime, endTime, result	Support future productivity and traceability analytics
SafetyEvent	eventType, severity, source, timestamp, resolutionStatus	Record warnings, exceptions and safety-related events
HMICommand	commandType, source, acknowledged, timestamp	Record high-level operator commands sent from HMI/DHMI

A ROS 2-to-NGSI-LD bridge will subscribe to posture, table, robot and mission topics and update the corresponding Orion-LD entities through NGSI-LD operations. A reverse bridge or NGSI-LD subscription can later publish context changes into ROS 2 where required. Raw camera frames should not be transferred to the context layer; only derived, non-identifying information necessary for ergonomic assessment and system operation should be shared.

With all of this explanation, in the next section we will show the logic of how this works together, and the operator can actually use this system.

2.4 Logic flow-chart showing how does the operator interact with the system

⇓ Operator selects a profile or gives a command through the HMI/DHMI
⇓ HMI/DHMI validates the command and maps it to a high-level ROS 2 intent
⇓ ROS 2 Vulcanexus routes the intent to the corresponding module
⇓ Ergonomic table node ErgoAssist/UR node Camera/HRI node
⇓ Arduino -> sensor/relays UR driver -> robot perception data
⇓ status, measurements, events
⇓ Optional monitoring bridge -> NGSI-LD/Orion-LD -> dashboard/reports

This is also better showcased in the Annex.
The idea is that always, as the Ethics Assessment indicated, the operator remains able to pause or stop the operation, while variables are provided to them to make their working conditions better so they do not have to bend to the robot, and robot-human is enhanced.

Moreover, when the voice command is not recognised, the HMI must show a fallback control. If a sensor, camera or robot error occurs, the relevant module must publish an exception state and the system must move to a defined safe condition.

2.5 Integration with other systems

The integration achieved so far is at component and subsystem level.
ROS 2 Vulcanexus has been used as the common communication environment for the Arduino-controlled table, the web camera and the Universal Robot. The Arduino/table tests confirmed the exchange of sensor and actuator-related information. The UR tests confirmed that Vulcanexus can communicate with the robot through the UR driver and that commands can be sent both interactively from a terminal and through scripts.

The camera/ROS4HRI perception pipeline, particularly face detection and 6D-frame detection, has been identified as a source of system instability in the current PoC setup. In our tests, enabling these components can crash the perception node and, in some configurations, affect the stability of the broader ROS 2 environment. To address this, the architecture isolates the `Posture-estimation node` as a containerised or separate process, communicating with the rest of the system only through typed ROS 2 interfaces (`/posture/observation`, `/posture/keypoints`, `/posture/joint_angles`, `/posture/risk_level`). A diagnostics/lifecycle/watchdog component monitors the health of the perception node and enforces timeouts and safe fallback, so that a perception crash does not propagate to the `ErgoAssist / mission coordinator` or the hardware-control nodes.

As an immediate technical priority, we are:

- Contacting the ROS4HRI maintainers (L. Ferrini and F. Gebelli) to discuss known compatibility issues and recommended dependency configurations for face and 6D-frame detection in ROS 2 Humble / Vulcanexus.

- Evaluating alternative 2D/3D pose-estimation implementations (e.g. MediaPipe- or YOLO-Pose-based nodes) that publish onto standard ROS 2 body/posture topics and can be run in an isolated container.

- Pinning compatible versions of ROS4HRI, OpenCV and related libraries for the stable subsystems (camera driver, UR3 driver, Arduino gateway), while keeping the perception node isolated until its dependencies are fully validated.

This approach aligns with the reviewer's recommendation to resolve library dependency conflicts within Vulcanexus or to containerize a stabilized 2D/3D pose node publishing onto standard ROS 2 body topics.

The planned context layer uses Orion-LD as an NGSI-LD-compatible Context Broker. It provides a shared semantic representation of the workstation for the HMI, analytics, logging and future integration with FIWARE-compatible systems. This layer has been designed but has not yet been tested experimentally; the current D2.1 should not state that Orion-LD dashboards or the ROS 2–Orion-LD bridge are already operational.

The initial NGSI-LD entities are as follows:

- OperatorPosture: keypoints, jointAngles, riskLevel, observedAt.
- ErgonomicTable: height, distanceRaw, state, targetHeight, observedAt.
- RobotStation: currentPose, robotMode, safetyStatus, activeMission, observedAt.
- SafetyEvent: eventType, severity, source, timestamp, resolutionStatus.
- HMICommand: commandType, source, acknowledged, timestamp.
- ProductionCycle: cycleId, SKU, startTime, endTime, result.

A ROS 2-to-NGSI-LD bridge will subscribe to posture, table, robot and mission topics and update the corresponding Orion-LD entities through NGSI-LD operations. A reverse bridge or NGSI-LD subscription can later publish context changes into ROS 2 where required. Raw camera frames should not be transferred to the context layer; only derived, non-identifying information necessary for ergonomic assessment and system operation should be shared.

FIWARE Context Broker integration is a core ARISE pillar and should not be postponed. Deploying an Orion-LD container and verifying live entity updates (in particular ErgonomicTable and RobotStation) via the ROS 2–NGSI-LD bridge is therefore defined as the very next technical step after finalising the ROS 2 interface contract and the voltage-to-distance calibration. The immediate work plan includes:

- Deploying a local Orion-LD instance (Docker container) and configuring the NGSI-LD endpoint.
- Implementing the ROS 2–NGSI-LD bridge node that maps selected ROS 2 topics to NGSI-LD entity attributes.
- Verifying live creation and update of entities such as ErgonomicTable (urn:posture:table:1) and RobotStation (urn:posture:ur3:1) with observed timestamps and consistent data types.
- Documenting the entity models, NGSI-LD payloads and basic latency and robustness tests.

This sequence ensures that the FIWARE context layer becomes operational early in the next phase steps, in line with the ARISE commitment to reusable, interoperable modules.

Ergonomic table and Sharp IR distance sensor

The ergonomic table is controlled via an Arduino Uno that switches two relays (UP / DOWN) emulating the original table buttons. An analog Sharp GP2Y0A21YK0F infrared distance sensor measures the perpendicular distance between the table surface and the floor, providing an indirect measurement of table height. The Arduino reads the sensor voltage and publishes it to ROS 2 via a micro-ROS or serial-to-ROS 2 gateway on topics such as /arduino_sensor (raw voltage) and subscribes to /arduino_cmd (high-level commands such as UP, DOWN, STOP) to activate the relays.

The current implementation provides raw sensor voltage; the next implementation step is to add a ROS 2 function that converts the measured voltage into a calibrated distance value (e.g. in millimetres) using a sensor-specific calibration curve. This calibrated distance will then be linked to the posture-aware logic, so that the measured table position (or an operator-related distance derived from it) can be used by the ergonomic-adjustment function. The calibration curve, units, valid operating range and error behaviour will be documented before the ergonomic KPI is evaluated.

The full characterisation of the Sharp infrared distance sensor, validated calibration curve, operating range, and robustness to ambient factory lighting (including verification that the station is shielded from direct sunlight), will be documented in D2.2, together with the operator feedback collected during the PoC validation sessions.

Moreover, we have started building out HMI tutorials but we need to do this for all the subcomponents that will be released as open-source, and we also want to make a case study with this particular need, and publish it, and present it at a robotics conference.

2.6 Human centricity

We have read the Initial Ethics Assessment. We have appointed Beatriz Zambrano, as the ethics representative, as she is part of Metamedics (the tech provider), and not the end user, if there were any concerns, the operators could go to her without being worried about their work.
Moreover, we have visited the plant, established an Advisory group of 8 workers and a leader. We have their written consent to participate. We have made interviews, and recorded measures. These measures have been anonymized and have not been shared with the company nor the workers. The data will not be used in any way by the company to monitor, or give any sort of scoring system to the workers to rank them as “efficient.” This logic is because as of right now, the work station does not adapt to them, the scores will not be very amicable, and we don´t want the workers to feel frustration without them being able to change their conditions. After the implementation and improvement, we will share

aggregated data with the workers. Additionally, the main purpose of the data is to construct a case study of how ergonomics improvements are good for the workers and for the company.

In this site visit,
1. We saw that the workers felt frustration with their work station, because they believed working with the robot made their work more difficult and slow. They had to stop the robot, and wait, whereas if they worked without the robot, the throughput would be better.
2. We also saw that the workers preferred simple visuals and commands. We made four tablet designs and the workers preferred the one that resembled their robotics UI/UX. This is a youtube video where you can see all the designs from the perspective of the workers and the leaders: https://www.youtube.com/watch?v=p1ySwrBxSxc
3. We are currently deciding not to use AI in the project because of the complexity of the EU AI Act. We are using simple voice-text and text-speech commands when the workers "command" the robot's angle, height etc to move. Additionally, because there is noice in the factory, and they have different accents etc, we will enable them to move these variables in the tablet by hand.

The main human-centred limitations currently identified are linked to the camera/HRI pipeline. Face detection and 6D-frame detection can crash the system, so these functions must not be used in an operator trial until their stability has been demonstrated. A manual or HMI-only fallback should remain available during validation. This was identified as a risk in the Initial Ethics Assessment, and we are taking a close look at it.

The ergonomic table is intended to support posture adaptation. At present, the sensor voltage is available and the table actuation functions have been tested, but the voltage-to-distance conversion and its linkage to the posture function remain incomplete. This is a direct priority because it is the step that transforms a hardware measurement into an ergonomic control variable.

Operator data will continue to be limited to what is necessary for the evaluation. The project will continue to distinguish raw camera information from derived, non-identifying posture metrics and should apply the agreed consent, pseudonymisation, retention and opt-out procedures before field validation.

On a final note, we have prepared an initial 10 minute presentation for this deliverable.
As we have been doing more component testing, we have not shown many humans, but we will in the next deliverable (D2.2), as we are aware that this should be done.

2.7.1 Stakeholders Mapping

Pepri Toys:

1. Operators (primary users, need reduced strain, control, no surveillance);
2. Line/production leads and Executive Sponsor at Pepri (adoption, throughput, plant integration);
3. Health & safety representatives (compliance, operator wellbeing, Advisory Group);

MMVR and University of Alicante (technology providers, delivery and implementation);
ARISE consortium (middleware interoperability, open assets);
Future SME adopters and integrators (replicability);
Works-council/worker representation (job security, fair use of monitoring data).

2.7.2 User Needs

- **Collection methods:** site-visit observations and structured interviews across shifts; the cross-shift Operator Advisory Group as a standing feedback body; HMI feedback

rounds currently running; think-aloud tests under realistic factory conditions (noise, gloves, time pressure) planned.

- **How input shapes design:** Advisory Group and interview findings become explicit requirements (the User Needs Document) governing HMI, ROS 2 interface contracts and mechanical design. If understanding of needs shifts during validation, a redesign sprint is triggered — SUS below threshold on any subgroup forces rework before the project advances.
- **Tasks to improve** → automation skills: manual holding of parts → robot present-and-hold at neutral posture; manual rotation → force-limited assisted rotation; awkward-angle visibility → height/tilt fixture auto-set from profile; interruptions/pace pressure → operator-controlled pause/nudge/resume; no ergonomic data → automated FIWARE logging and alerts.
- **Help**: with tutorials, github repositories and instructions, and video webinars.

2.7.3 Participation practices

Pepri operators have been asked to be involved already at M1, at the Site Lead, sampled for gender balance, age (20–55+), experience level, handedness and height range.
Activities in which they are asked to be involved include:

1. HMI feedback,
2. Think-aloud usability tests,
3. PoC validation sessions

Usability scores (SUS), HRI-trust scores, task/cycle data, non-identifying posture proxies, and acceptance survey responses as well as RULA/REBA, have all been recorded.

Anonymisation is ensured because only derived non-identifying metrics leave the edge; raw visual data is discarded after extracting anonymous skeletal coordinates; logs are pseudonymised with a 90-day retention limit, and of course participants are aware of what is collected and why, purpose limitation, opt-out rights, retention period, via a plain-language notice at the cell. Informed consent: explicit written consent before any data collection, with opt-out from analytics at any time without penalty or loss of cell access. The data collected has been done so taking into account compliance (full ISO 10218 / ISO/TS 15066)

2.8 Validation and Testing Tools

The PoC has used incremental functional testing of individual components and subsystems.
The tests performed include:

- ROS 2/Vulcanexus communication with the Arduino;
- acquisition of the distance-sensor voltage;
- relay activation and deactivation;
- motor activation in both directions;
- web-camera connection with Vulcanexus;
- UR-driver communication with the Universal Robot;
- direct robot commands sent from a terminal;
- execution of robot scripts containing movement instructions;
- factory observation and operator surveys for HMI and interaction requirements.

The results show successful component-level operation for the Arduino/table subsystem and the UR/Vulcanexus subsystem. The camera connection is functional at connection level, but the complete face/6D-frame processing chain is not yet resilient because of crashes and library compatibility issues.

We can provide here some evidence of this through google drive link to videos:

1. Screencast from 09-04-2026 06_55_43 AM.webm
2. Screencast from 09-04-2026 07_24_19 AM.webm
3. Screencast from 09-04-2026 07_48_49 AM.webm
4. Screencast from 09-04-2026 07_54_03 AM.webm
5. Screencast from 09-04-2026 07_55_03 AM.webm

The following tests remain pending:

- automated ROS 2 package and interface tests;
- formal validation of the voltage-to-distance calibration curve;
- validation of the posture function using the converted distance;
- controlled tests of robot pause, yield, resume and exception states;
- camera pipeline isolation and recovery tests;
- ROS4HRI compatibility testing with pinned library versions;
- voice-command tests in factory noise;
- ROS 2–Orion-LD latency and data-completeness tests;
- integrated safety validation before tests with operators.

We are setting standards of quality so that a test should be considered passed only when the result is reproducible, the software configuration is recorded, the expected output is defined, and the corresponding failure behaviour is documented.

2.9 Available Software Modules

Module	Open-Source link	Description	Standards Frameworks	Documentation Available
1. The inclusive touch screen HMI	Windows https://drive.google.com/file/d/16sf7bNk-7SvtmEH8Ivq5UkgBq_Mr3j2T/view?usp=drive_link	The inclusive touch-screen HMI is the point of contact between the operator and the POSTURE collaborative cell It runs on an industrial touch panel PC with no keyboard or mouse, and its purpose is to let any operato, regardless of language, visual acuity, handedness, height, prior experience, or glove us, control the robot safely through high-level intent commands, never joint movements.	Unity (Windows + Linux) Unity UI Toolkit (UXML + USS) Unity Input System Universal Render Pipeline C# / .NET (assembly definitions) ROS 2 Jazzy / Vulcanexus rosbridge v2 (rosbridge_suite) WebSocket (RFC 6455), TCP 9090 JSON over std_msgs/msg/String FIWARE NGSI-LD (ETSI B612 (SIL OFL) · Material Icons (Apache 2.0)	Tutorial https://www.youtube.com/watch?v=p1ySwrBxSxc

2. The base project for the XR micro-onbo ardings	https://github.com/ MetamedicsVR-org /Unity-VR-Robotica	This is a private repository, including the simulation base project, with log-in, analytics, scenes etc. We cannot make it public at this time	OpenXR Universal Render Pipeline 17.3.0 OpenXR Plugin 1.16.1 Unity OpenXR Meta 2.5.0, XR Interaction Toolkit 3.4.1 XR Hands 1.7.3 Input System 1.19.0 Architecture ARM64, Backend scripting IL2CPP API graphic Vulkan	https://bootcamp -vr.metamedicsvr. com/en/bootcamp

Additionally,

Arduino/table control	Available for component testing	ROS 2/Vulcanexus communication with Arduino; sensor voltage reading; relay and motor control in both directions	ROS 2, Vulcanexus, Arduino I/O	Implement voltage-to-distan ce conversion and connect it to posture logic
Universal Robot control	Available for functional testing	UR driver configured with Vulcanexus; tested through direct terminal commands and scripts	ROS 2, Vulcanexus, Universal Robots driver	Integrate formal ErgoAssist actions and complete safety/exception validation
Web-camera connection	Available at connection level	Camera connected and tested with Vulcanexus	ROS 2/Vulcanexus, camera driver	Stabilise face and 6D-frame processing
ROS4HRI/perc eption layer	Experimental/WIP	Evaluated as the intended HRI and perception layer	ROS4HRI, ROS 2/Vulcanexus	Library compatibility problems and crashes remain unresolved
ErgoAssist skill	Partially available/WIP	Intended to expose reusable actions for presentation, holding, rotation and yield-to-human	ROS 2 actions/services, Vulcanexus	Formalise interfaces and connect to validated UR behaviours
FIWARE/NGSI-LD bridge	Not yet available as tested module	Planned mapping of ROS 2 events to Orion-LD context entities	FIWARE, Orion-LD, NGSI-LD	Define data models and perform first component test

3 PoC Assessment

3.1 Technical and Business KPIs

KPIs					
KPI (Description)	**Type**	**M1 value**	**PoC value**	**M12 value**	**KPI (Description)**
RULA/REBA score on targeted subtasks	Technical (Ergonomics)	~5–6 (medium-high risk)	Baseline captured during site visit; PoC reduction not yet measured	≤4 (≥1-level reduction)	RULA/REBA score on targeted subtasks
NASA-TLX overall score	Technical (Human Factors)	~55–65 (est. repetitive precision assembly)	Not yet measured (validation from M4)	≤47 (–15%)	NASA-TLX overall score
Cycles per hour	Business	To be set from site-visit baseline data	Not yet measured	+10%	Cycles per hour
% operators preferring the new cell	Business	0% (pre-deployment)	Not yet measured (post-validation)	≥80%	% operators preferring the new cell
% operators identifying user-friendliness	Technical (Usability UX)	0% (pre-deployment)	HMI in feedback rounds with Advisory Group; formal SUS pending	≥80%	% operators identifying user-friendliness

3.2 Communication and Dissemination KPIs

Activity	Activity performed in this period	Target KPI
Promotion of project outcomes via posts on ARISE website and partner social media (MMVR, UA, Pepri)	Post in LinkedIn	≥1 per milestone (4 total)

Project presentations at industrial HRI / robotics conferences	None yet (planned from M10)	Minimum of 2
Peer-reviewed scientific publications	None yet (planned M10–M12)	2 open-access by M12
Partner enablement tutorials and integration guides	HMI video tutorial + document available	≥1 tutorial video; ≥4 XR tutorials; 1 integration guide by M12

Additionally, we have built in other internal KPIs that we are using internally between technical developers to ensure the coordination functions well, and all the parts are tested as a component, and as a unit.

KPI	Type	Initial/current status	PoC target or next measurement
Arduino–ROS 2 communication	Technical	Functionally tested	Reproducible test with logged messages and error handling
Distance-sensor voltage acquisition	Technical	Functionally tested	Calibrated voltage-to-distance conversion with documented error
Relay activation and motor direction	Technical	Both directions tested	Repeatability and safe stop validation
UR driver–Vulcanexus–robot communication	Technical	Successfully tested	Integrate formal actions and state feedback
Robot command methods	Technical	Terminal commands and scripts both successful	Compare repeatability and document supported method
Camera connection	Technical	Connection tested	Stable frame acquisition under intended conditions
Face/6D-frame pipeline	Technical/HRI	Not yet passed; crashes observed	No crash during defined test duration and controlled recovery
ROS4HRI compatibility	Technical	Not yet resolved	Pinned compatible environment or justified alternative

Voltage-to-distance conversion	Technical/ergonomics	Pending	Validated calibration curve and known operating range
Posture-function integration	Technical/ergonomics	Pending	Distance measurement correctly influences posture-related logic
ROS 2–FIWARE latency	Technical	Not measured; Orion-LD test pending	Target below 2 seconds after bridge implementation
Cycles per hour	Business	Not yet measured in the PoC	Compare against the Pepri baseline without reducing safety or quality
Operator acceptance	Business/human factors	Qualitative feedback collected	Quantify acceptance and identify subgroup differences
Safety events and unsafe contacts	Safety	Formal integrated validation pending	Zero unsafe contacts in the defined test scenarios

3.3 Industrial added value through human-robot interaction

POSTURE addresses a concrete, costly problem at the Pepri station: operators manually hold, rotate and position blow-moulded parts during high-frequency cycles, driving sustained wrist, neck and shoulder strain and generating operator complaints, with no ergonomic data currently collected. The HRI implementation transfers the strain-heavy holds and rotations to a collaborative robot while keeping the operator in control of pace and process — the value comes specifically from the human-robot interaction design, not from automation replacing the operator.

Expected impact at the end user (Pepri). The core skill (ErgoAssist) presents and holds parts at each operator's neutral posture and standardises low-effort micro-adjustments. This is expected to deliver at least one level of RULA/REBA improvement on the targeted subtasks, a ~15% reduction in perceived workload (NASA-TLX), fewer near-misses, and a 5–10% throughput increase. Against 3,000–4,500 productive hours per station per year, these gains translate into a measurable, trackable ROI against a bounded deployment cost, with payback in the single-station range. Critically, the capacity freed by reduced injuries and interruptions is committed to quality-improvement work rather than headcount reduction, which is what makes the intervention acceptable to operators and their representatives.

Why the HRI approach drives the value. Very few existing cobot deployments in packaging offer posture-aware primitives that adapt to individual operator profiles, human-in-the-loop exception handling based on pause–nudge–resume rather than full E-stop and reteach, or built-in ergonomics analytics connected to enterprise systems via open standards. POSTURE combines these: per-operator preset management, force-limited rotation that removes biomechanically harmful grip patterns, and a real-time FIWARE monitoring layer that makes ergonomics data auditable and portable. The operator-controlled pace and interruptibility are what convert a standard cobot cell into an ergonomics-first, human-centred one.

Replicability and ecosystem value. The value is designed to travel. The ErgoAssist ROS 2 skill, portable NGSI-LD data models, and thin REST/CSV connectors are built for reuse across plants with different robots, stations and SKUs. Published as an open reference implementation in the ARISE catalog — with clear ROI figures and an HRI inclusivity and ergonomics checklist — the work is intended to shorten procurement cycles and de-risk trials for other European SMEs facing the same ergonomic challenges. The Pepri deployment becomes the anchor case study for that wider adoption.

Status. At mid-M3, the industrial impact above is expected rather than demonstrated: the human-centricity groundwork (site visit, baseline ergonomic data, Advisory Group, HMI) is in place, and the quantified gains will be measured during PoC validation from M4 onward.

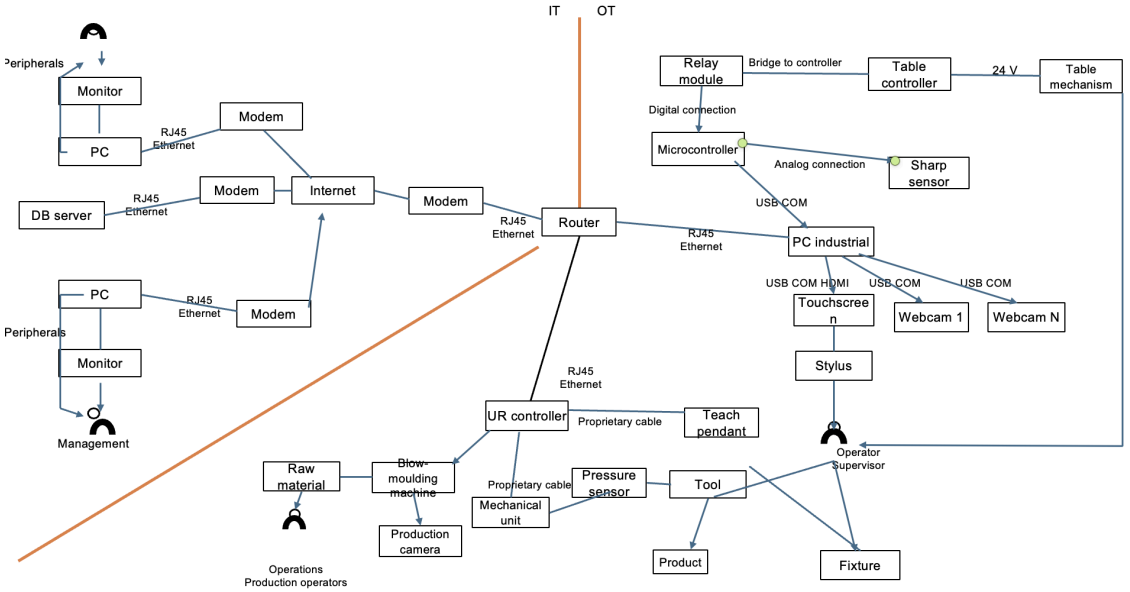
3.4 Risk evaluation and mitigation plan

Barrier/Challenge	Action to mitigate the risk	Lessons learnt
Interface contract (ROS 2 spec and division of tasks)	Interface contract is a formal M1 deliverable (T3.1); an ErgoAssist stub/simulator lets MMVR do offline integration testing progress tracked every bi-weekly sprint	The project handbook helped UA and MMVR understand which parts were going to be done by who, in which timing and why. Then, from this document there will be a data contract for specific components
Station hardware definition	MMVR builds an HMI version for internal testing from M1, independent of the hardware decision; hardware spec is a formal T2.2 output (M3); escalate to steering committee if unconfirmed	Great to start with subcomponent development and testing early, even if things have to be changed then later when unifying it.
ROS 2 → FIWARE latency exceeding 2-second threshold	Latency measured from first integrated dry-run (M3) and tracked as a primary KPI; network config validated in T2.2; Orion-LD bridge / edge buffering as fallback	Not yet encountered — no integration run performed at mid-M3, priority for d2.2.
Operator recruitment and scheduling constraints	Advisory Group established M1 with H&S involvement; Site Lead formally owns scheduling; test sessions ≤45 min, runnable during shift transitions	Recruited both: men and women, both: young and older, both: left-handed and right-handed, both: nationals and non nationals to ensure diversity!
Baseline data gaps after site visit	Baseline ergonomic data gathered on target tasks; Meta Quest 3 LiDAR scene mesh + 360° video reconstruct the cell as a digital twin for remote offline measurement	

4 Annex

(1) POSTURE Diagram

The intention of this diagram is to show the physical layout of the POSTURE workstation, split into an IT side (office PCs, servers, and management connecting through modems to the router) and an OT side (the shop-floor machines), separated by the orange IT/OT boundary. On the OT side, the industrial PC acts as the hub linking the touchscreen HMI, webcams, and microcontroller, which drives the adjustable ergonomic table (relay, controller, Sharp distance sensor), while the UR robot controller runs the cell hardware (blow-moulding machine, tool, sensors, fixture) around the product. The operator/supervisor sits at the center of the robot cell, showing that the human stays in control of the process throughout.



(2) Intended TEF Diagram

Within the ARISE design-for-reuse philosophy, **ErgoAssist** is the primary skill intended for publication as a reusable module. The remaining skills shown are supporting components of the POSTURE cell; some are still under development and their release as standalone reusable modules will be assessed as they mature.

